

Fangyuan Wang

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Final-year Ph.D. candidate at The Hong Kong Polytechnic University researching robot learning and embodied AI. Develops vision-language-action models, language-grounded planning, and reinforcement learning methods for long-horizon manipulation, with experience spanning simulation and real-robot evaluation.

Education

The Hong Kong Polytechnic University Ph.D. candidate in Mechanical Engineering; advisor: Prof. David Navarro-Alarcon.	Sep 2022 – Present
National University of Singapore Visiting Student; advisor: Prof. Guillaume Sartoretti.	Sep 2025 – Mar 2026
Zhejiang Sci-Tech University Master's degree in Software Engineering; advisor: Prof. Jian Hou. National Scholarship.	Sep 2019 – Jun 2022
Zhejiang Sci-Tech University B.Eng. in Computer Science & Technology.	Sep 2014 – Jun 2018

Research Experience

Vision-Language-Action Models & Long-Horizon Manipulation Advisor: Prof. David Navarro-Alarcon, The Hong Kong Polytechnic University	Sep 2022 – Present
ThinkProprio (CoRL 2026, accepted): Developed proprioception- and instruction-guided visual token selection before VLM computation. On CALVIN ABC→D, retained 12% of visual tokens, reduced inference latency from 52 to 22 ms versus FLOWER on an RTX 4090, and improved average task-chain length from 4.44 to 4.52. Evaluated on CALVIN, LIBERO, and real-world manipulation. IALP (AAAI 2025, oral): Developed a closed-loop planning system that combines multimodal perception and LLM reasoning to ground natural-language instructions into executable PDDL plans for mobile manipulation. EISP (T-ASE 2025): Developed explicit-implicit subgoal planning with hindsight sampling and value-based selection to generate feasible subgoal sequences for sparse-reward, long-horizon tasks; evaluated in simulation and on real robots.	
Human-Like Behavior Generation for Humanoid Robots Advisor: Prof. David Navarro-Alarcon, The Hong Kong Polytechnic University	Sep 2024 – Present
Developed HuBE , a morphology-aware behavior generation framework conditioned on robot state, target poses, and situational context. Constructed the HPose dataset and used bone-scaling augmentation for cross-embodiment transfer; evaluated generated behaviors on humanoid platforms, including real-robot execution on GR1.	
Byzantine-Resilient Decentralized Learning Advisor: Prof. Jian Hou, Zhejiang Sci-Tech University	Oct 2021 – Sep 2022
Designed credibility-based parameter aggregation using immediate feedback and historical node reliability to suppress adversarial updates in decentralized learning; evaluated robustness on MNIST and CIFAR-10.	
Active Flow Control & Sensor Selection Visiting research with Prof. Ning Gui, Central South University	Aug 2021 – Sep 2022
Developed an attention-based feature-selection framework for reinforcement learning control. Selected 5 sensors from 151 candidates for two-dimensional Kármán vortex-street flow control, reducing sensing requirements while preserving effective drag reduction.	
Multi-Agent Resilient Consensus Advisor: Prof. Jian Hou, Zhejiang Sci-Tech University	Jul 2020 – Jun 2021
Developed DDPG-based adaptive neighbor weighting and the neural-network-free Q-consensus law to mitigate adversarial agents while maintaining convergence; demonstrated resilient speed synchronization on a six-vehicle platform.	

Selected Publications

* Equal contribution † First student author (supervisor is first author)

- [1] **F. Wang**, P. Zhou, J. Qi, S. Lyu, C. He, D. Navarro-Alarcon, and G. Guo, “Think Proprioceptively: State-Grounded Visual Token Selection for VLA Policies,” *Conference on Robot Learning (CoRL)*, 2026. **Accepted**.
- [2] **F. Wang**, Z. Wang, G. Pei, et al., “World Models for Robotic Manipulation: A Survey,” *SmartBot*, Early View, e70053, 2026. doi:10.1002/smb2.70053.
- [3] **F. Wang**, et al., “Instruction-Augmented Long-Horizon Planning: Embedding Grounding Mechanisms in Embodied Mobile Manipulation,” in *Proc. 39th AAAI Conference on Artificial Intelligence*, 2025, pp. 14690–14698. **(Oral Presentation)**
- [4] **F. Wang**, et al., “Explicit-Implicit Subgoal Planning for Long-Horizon Tasks With Sparse Rewards,” *IEEE Transactions on Automation Science and Engineering*, vol. 22, pp. 16038–16049, 2025.
- [5] S. Lyu, **F. Wang**, et al., “Cross-embodiment human-like behavior execution for humanoid robots,” *Robot Learning*, vol. 3, no. 2, Art. no. 0015, 2026.
- [6] S. Lyu, **F. Wang**, W. Lin, G. Guo, and D. Navarro-Alarcon, “Enhancing End-user Engagement in Human-Robot Interaction by Performing LLM-driven Expressive Behaviors,” *ACM Transactions on Human-Robot Interaction*, in press, 2026.
- [7] J. Qi, L. Lu, **F. Wang**, et al., “LLM-driven Symbolic Planning and Hierarchical Imitation Learning for Long-Horizon Deformable Object Assembly,” *Robotics and Computer-Integrated Manufacturing*, vol. 97, 103096, 2026.
- [8] G. Gong, F. Hu, **F. Wang**, et al., “TEVIO: Thermal-Aided Event-Based Visual-Inertial Odometry for Robust State Estimation in Challenging Environments,” *IEEE Transactions on Instrumentation and Measurement*, vol. 74, Art. no. 7505211, pp. 1–11, 2025.
- [9] S. Huo, F. Hu, **F. Wang**, et al., “Rearranging Deformable Linear Objects for Implicit Goals with Self-Supervised Planning and Control,” *Advanced Intelligent Systems*, vol. 7, no. 2, 2400330, 2025.
- [10] J. Wei, Z. Qiu, **F. Wang**, W. Lin, N. Gui, and W. Gui, “Understanding via Exploration: Discovery of Interpretable Features With Deep Reinforcement Learning,” *IEEE Transactions on Neural Networks and Learning Systems*, vol. 35, no. 2, pp. 1696–1707, 2024.
- [11] J. Hou, **F. Wang**[†], et al., “Credibility Assessment Based Byzantine-Resilient Decentralized Learning,” *IEEE Transactions on Dependable and Secure Computing*, vol. 20, no. 5, pp. 3734–3746, 2023.
- [12] J. Wei*, **F. Wang***, et al., “An Embedded Feature Selection Framework for Control,” in *Proc. 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2022, pp. 1979–1988.
- [13] J. Hou, **F. Wang**[†], L. Wang, and Z. Chen, “Reinforcement Learning Based Multi-Agent Resilient Control: From Deep Neural Networks to an Adaptive Law,” in *Proc. 35th AAAI Conference on Artificial Intelligence*, 2021, pp. 7737–7745.

Under Review

- [14] S. Lyu, L. Sheng, **F. Wang**, et al., “Phy-Tac: Toward Human-Like Grasping via Physics-Conditioned Tactile Goals,” under review.

Technical Skills

Development: Python, PyTorch, Weights & Biases; remote HPC experimentation.

Methods: VLA models, reinforcement learning, imitation learning, LLM-based task planning, PDDL.

Robotics: Long-horizon and mobile manipulation, humanoid behavior generation, simulation and real-robot evaluation.

Awards & Academic Service

Champion, Domestic Robotics Track

Oct 2025

2nd Zhuhai International Dexterous Manipulation Challenge (team award).

Reviewer: TNNLS, T-RO, T-ASE, AAAI, ICLR, ICML, ICRA, IROS.